HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • **Stamp 1G — Full-Time Work Eligible**

PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Translated stakeholder needs and technical constraints into **system-level requirements** and design inputs for medical devices, ensuring functional coverage and traceability across subsystems.
- Led technical problem-solving initiatives using a structured, data-driven systems engineering approach to resolve complex issues within the product lifecycle.
- Acted as a system owner and subject matter expert, taking ownership of technical decisions impacting the overall medical device architecture and performance.
- Supported Verification & Validation, Risk Management, and Quality Management System (QMS) activities, working within established medical device design controls.
- Collaborated with cross-functional teams including Engineering, Quality, Regulatory Affairs, and Clinical to define system architecture and technical solutions.
- Facilitated architectural and design decisions to support the successful delivery of technical milestones and project objectives.
- Containerized **Java Spring Boot** microservices using **Docker** and deployed them to **AWS EC2** instances, managing CI/CD pipelines with **Jenkins**.

SKILLS

Systems Engineering – System Requirements, Requirements Decomposition, Traceability, System Architecture, System Integration, Design Inputs, Risk Management, Verification & Validation, Quality Management Systems (QMS)

Medical Devices – MedTech Product Development Lifecycle, Design Controls, ISO 13485, ISO 14971

Programming Languages – Java, Python, SQL

Frameworks & Libraries – Spring Boot, pandas, NumPy, scikit-learn

Cloud & DevOps – AWS (EC2, S3, Lambda, CloudWatch), Docker, Jenkins, Git

Databases – PostgreSQL, MySQL, Oracle

Tools – Splunk, Dynatrace, Jira, Confluence, Power BI, Tableau

PROJECTS

Medical Device System Requirements Specification

- Defined and documented comprehensive system-level requirements for a new medical device, translating user needs and regulatory constraints into actionable design inputs.
- Decomposed high-level system requirements into detailed subsystem specifications for hardware, firmware, and software components, ensuring complete functional coverage.
- Developed traceability matrices linking requirements to design elements and V&V; test cases, supporting the medical device design control process.

Automated Production Monitoring Dashboard

- Developed a **Python**-based data pipeline to ingest operational metrics from **AWS CloudWatch** and **Splunk** logs, feeding into a **Power BI** dashboard.
- Engineered **SQL** queries across **PostgreSQL** and **MySQL** to aggregate transaction data and identify performance bottlenecks, contributing to a **15%** reduction in incident response time.
- Designed and implemented REST APIs using **Java Spring Boot** to expose key system health indicators for integration with enterprise monitoring tools.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)